Practical No. 5A

Aim: Programs on Menus and Adapters. (Spinner, ListView, Menu)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:tools="http://schemas.android.com/tools"
   android:layout_height="match_parent"
   android:layout_width="match_parent"
   xmlns:android="http://schemas.android.com/apk/res/android">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/tvSpinner"
      android:text="Spinner"
      android:textSize="35dp"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="30dp" />

   <Spinner
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/spinner1"
      android:layout_marginRight="20dp"
      android:layout_below="@id/tvSpinner"/>

   <Spinner
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/spinner2"
      android:layout_toRightOf="@+id/spinner1"
      android:layout_below="@id/tvSpinner"/>

   <Button
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/btn_click"
      android:text="Click Here"
      android:layout_below="@id/spinner1"
      android:textSize="20dp"/>

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/tvSpinnerOutput"
      android:layout_below="@id/btn_click"
      android:textSize="20dp"/>
```

```
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tvListView"
        android:text="ListView"
        android:textSize="35dp"
        android:layout_centerHorizontal="true"
        android:layout_below="@id/tvSpinnerOutput"
        android:layout_marginTop="30dp" />

    <ListView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/listView"
        android:layout_below="@+id/tvListView"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tvListViewOutput"
        android:layout_below="@id/listView"
        android:textSize="20dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tvMenu"
        android:text="Menu"
        android:textSize="35dp"
        android:layout_centerHorizontal="true"
        android:layout_below="@id/tvListViewOutput"
        android:layout_marginTop="30dp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tvMenuOutput"
        android:layout_below="@id/tvMenu"
        android:textSize="20dp"/>

</RelativeLayout>
```

strings.xml

```
<resources>
    <string name="app_name">My Application</string>
    <string-array name="country_name">
        <item>India</item>
        <item>South Afrcia</item>
        <item>Brazil</item>
        <item>New Zealand</item>
```

```
        <item>Australia</item>
    </string-array>
</resources>
```

mymenu.xml

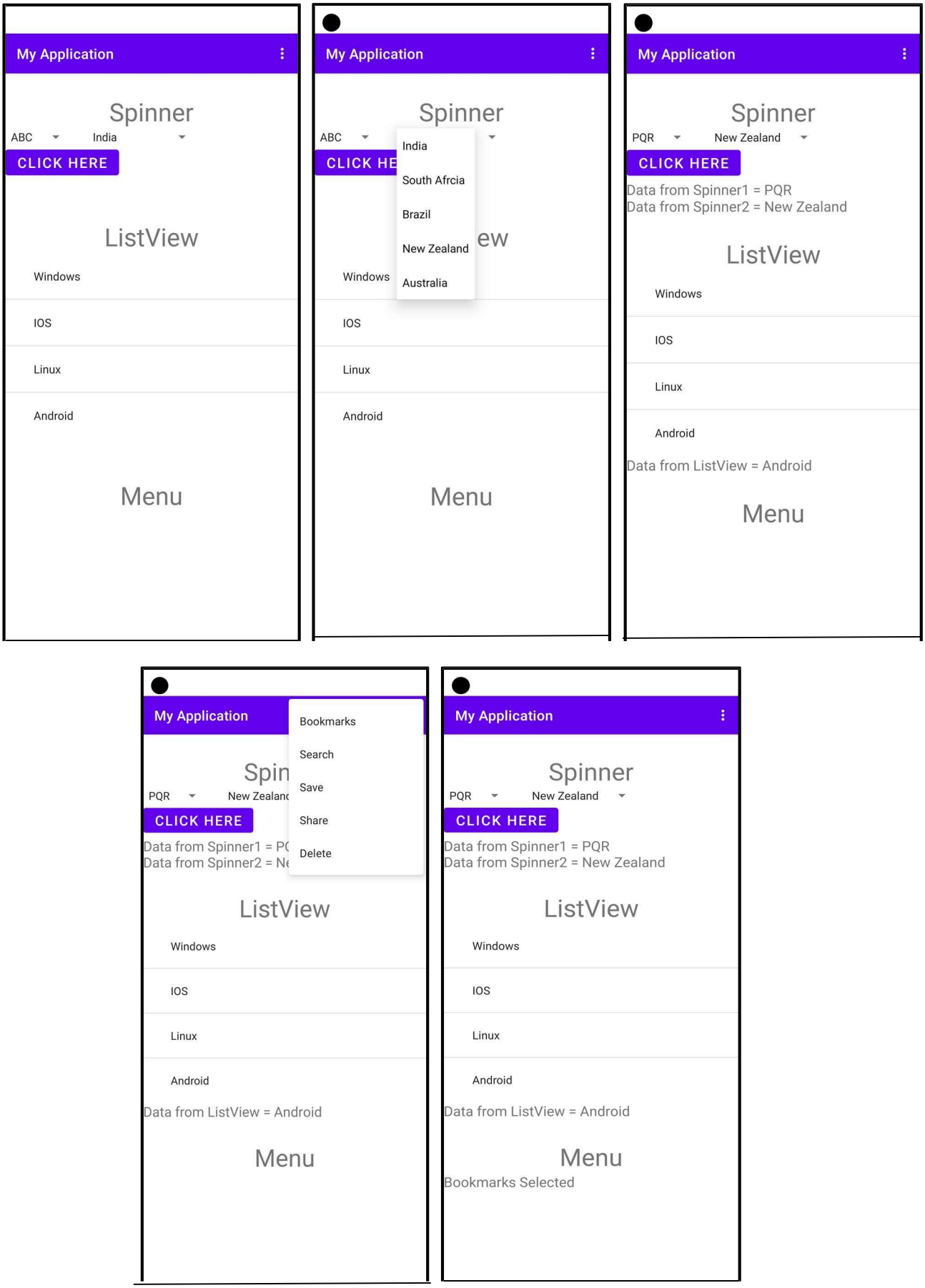
```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/bm"
        android:title="Bookmarks"/>
    <item
        android:id="@+id/sr"
        android:title="Search"/>
    <item
        android:id="@+id/sa"
        android:title="Save"/>
    <item
        android:id="@+id/sh"
        android:title="Share"/>
    <item
        android:id="@+id/dlt"
        android:title="Delete"/>
</menu>
```

MainActivity.kt

```
package com.example.lifecycle

import android.annotation.SuppressLint
import android.os.Bundle
import android.view.Menu
import android.view.MenuInflater
import android.view.MenuItem
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.get

class MainActivity : AppCompatActivity() {

    @SuppressLint("MissingInflatedId")
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val sp1 = findViewById<Spinner>(R.id.spinner1)
        val sp2 = findViewById<Spinner>(R.id.spinner2)
        val btnClick = findViewById<Button>(R.id.btn_click)
        val tvSpinnerOutput = findViewById<TextView>(R.id.tvSpinnerOutput)
        val lv = findViewById<ListView>(R.id.listView)
        val tvListViewOutput = findViewById<TextView>(R.id.tvListViewOutput)
```

```
        val lst = ArrayList<String>()
        lst.add("ABC")
        lst.add("PQR")
        lst.add("XYZ")

        val os = arrayOf<String>("Windows","IOS","Linux","Android")

        val adp1 = ArrayAdapter(this, android.R.layout.simple_spinner_item,lst)
        adp1.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)
        sp1.adapter = adp1

        val adp2 = ArrayAdapter.createFromResource(this,
R.array.country_name,android.R.layout.simple_spinner_item)
        adp2.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item)
        sp2.adapter = adp2

        val adp3 = ArrayAdapter(this, android.R.layout.simple_expandable_list_item_1,os)
        lv.adapter = adp3

        btnClick.setOnClickListener {
            val str1 = sp1.selectedItem.toString()
            val str2 = sp2.selectedItem.toString()
            tvSpinnerOutput.text = "Data from Spinner1 = $str1 \nData from Spinner2 =
$str2"
        }

        lv.setOnItemClickListener { adapterView, view, i, l ->
            val str3 = adapterView.getItemAtPosition(i) as String
            tvListViewOutput.text = "Data from ListView = $str3"
        }
    }

    override fun onCreateOptionsMenu(menu: Menu?): Boolean {
        menuInflater.inflate(R.menu.mymenu,menu)
        return super.onCreateOptionsMenu(menu)
    }

    override fun onOptionsItemSelected(item: MenuItem): Boolean {
        val tvMenuOutput = findViewById<TextView>(R.id.tvMenuOutput)
        when (item.itemId){
            R.id.bm -> tvMenuOutput.text = "Bookmarks Selected"
            R.id.sr -> tvMenuOutput.text = "Search Selected"
            R.id.sa -> tvMenuOutput.text = "Save Selected"
            R.id.sh -> tvMenuOutput.text = "Share Selected"
            R.id.dlt -> tvMenuOutput.text = "Delete Selected"
        }
        return super.onOptionsItemSelected(item)
    }
}
```

Output


My Application
Spinner
ABC
India
CLICK HERE
ListView
Windows
IOS
Linux
Android
Menu
My Application
Spinner
ABC
India
South Afrcia
Brazil
New Zealand
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PQR
New Zealand
CLICK HERE
Data from Spinner1 = PQR
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Bookmarks
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Save
Share
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Data from ListView = Android
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Bookmarks Selected